

DATA ANALYSIS WORKFLOW & GATHERING DATA

DATA WORKFLOW OVERVIEW



DATA COLLECTION STRATEGY



DATA SOURCES AVAILABLE AT BAYES

Data Type	Data Source	Content
Financial	Compustat (North America & Global)	Balance sheet, income statement, cash flow data
Financial	Capital IQ: Capital Structure	Detailed capital structure information
Market	CRSP (also, CRSP-Compustat Merged)	Stock prices, returns, trading volume
Market	Datastream	Global equities, economics, commodities data
Banking	Bank Regulatory Database	Bank structure, call reports, holding company financials

...

EXAMPLE: RESEARCH ON ESG AND FIRM VALUE

- **Research Question:**
 - *Does superior ESG performance increase firm value for listed companies?*
- **Key Variables:**
 - **Dependent Variable:** **Tobin's q** as a proxy for firm value
 - **Independent Variable:** **ESG score**
 - **Control Variables:** Firm size, leverage, profitability, etc.
- **Sample:**
 - Listed companies from **CRSP-Compustat Merged** database
 - Merged with ESG data in the dataset from Ch. 5 of <https://www.greenfinance.education>

TOBIN'S q AND ESG SCORE

- What is Tobin's q ?
- What is ESG?

When you do not understand something, AI is your friend!

DATA PREPARATION & CLEANING (BEFORE ANALYSIS)

- Initial assessment:
 - *Summary statistics* for all variables
 - *Missing data* patterns and outlier identification
- Common cleaning procedures:
 - Outlier treatment:
 - e.g., *winsorization*: Typically at 1% and 99%
 - Variable transformation:
 - e.g., *logarithmic* ($\log$): Turn a skewed distribution close to normally distributed

MODEL DEVELOPMENT

EXAMPLE: REGRESSION MODELS

- Pooled OLS regression:

- Basic approach combining *cross-sectional and time-series* data
- Example:

$$q_{i,t} = \alpha + \beta_1 ESGscore_{i,t} + \beta_2 Controls_{i,t} + \varepsilon_{i,t}$$

- Year and industry indicators:

- Control for *time trends and industry differences*
- e.g., $q_{i,t} = \alpha + \beta_1 ESGscore_{i,t} + \beta_2 Controls_{i,t}$

$$+ \beta_3 \sum Year_t + \beta_4 \sum Industry_i + \varepsilon_{i,t}$$

WHAT ARE YEAR AND INDUSTRY INDICATORS USED IN REGRESSION?

- Ask Grok Fast!

ANALYSIS & RESULTS

EXAMPLE: DESCRIPTIVE STATISTICS TABLE

Variable	N	Mean	Median	Std. Dev.	Min	Max
q	4,200	1.82	1.56	0.94	0.78	4.65
ESGscore	4,200	65.34	67.00	18.45	12.00	95.00
Firm_Size	4,200	8.45	8.32	1.75	4.62	12.85
Leverage	4,200	0.42	0.39	0.21	0.04	0.93

HOW TO PRODUCE A DESCRIPTIVE STATISTICS TABLE

ANALYSIS & RESULTS EXAMPLE: DESCRIPTIVE STATISTICS TABLE

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- Using Python
 - Ask **Qwen!**

ANALYSIS & RESULTS

EXAMPLE: CORRELATION TABLE

	q	ESGscore	Firm_Size	Leverage
q	1.000			
ESGscore	0.285***	1.000		
Firm_Size	0.124**	0.356***	1.000	
Leverage	-0.142**	0.034	0.289***	1.000

- Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

HOW TO PRODUCE A CORRELATION TABLE

- Using Python
 - Ask **Gemini 3 Flash** on **GitHub Copilot**

ANALYSIS & RESULTS EXAMPLE: CORRELATION TABLE

	<i>q</i>	ESGscore	Firm_Size	Leverage
<i>q</i>	1.000			
ESGscore	0.285***	1.000		
Firm_Size	0.124**	0.356***	1.000	
Leverage	-0.142**	0.034	0.289***	1.000

• Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

ANALYSIS & RESULTS

EXAMPLE: REGRESSION TABLE

Variable	Model 1	Model 2	Model 3	Model 4
ESGscore	0.027*** (0.008)	0.024*** (0.007)	0.022*** (0.007)	0.021*** (0.007)
Firm_Size		-0.128*** (0.035)	-0.132*** (0.034)	-0.135*** (0.034)
Leverage		-0.745* (0.371)	-0.732*** (0.154)	-0.728** (0.236)
Industry Indicators	No	No	Yes	Yes
Year Indicators	No	No	No	Yes
Observations	4,200	4,200	4,200	4,200
R-squared	0.081	0.298	0.325	0.337

- Note: Standard errors in parentheses; *** p<0.01, ** p<0.05, * p<0.1

ANALYSIS & RESULTS

EXAMPLE: ROBUSTNESS CHECKS

Variable	Main Results	Alternative ESG Measure	Subsample: Large Firms	Subsample: 2010-2019
ESGscore	0.021*** (0.007)	0.018** (0.008)	0.025*** (0.009)	0.019** (0.008)
Controls	Yes	Yes	Yes	Yes
Industry Indicators	Yes	Yes	Yes	Yes
Year Indicators	Yes	Yes	Yes	Yes
Observations	4,200	4,050	2,340	2,100
R-squared	0.337	0.328	0.356	0.342

- Note: Standard errors in parentheses; *** p<0.01, ** p<0.05, * p<0.1

GATHERING UNSTRUCTURED DATA

UNSTRUCTURED VS STRUCTURED DATA

- **Structured data**

- follows a consistent pattern and thus can be organized in a fixed format,
 - referred to as a **schema** that defines each field's data type (e.g., integer, text string, double[-precision]/floating-point, date, etc)
- Example:

ID	Name	Height (ft.)	Email	Join_Date
1324	John D	6.1	john@email.com	2024-01-01
2846	Mary S	5.5	mary@email.com	2024-01-15

- **Unstructured data**

- is inconvenient to organize into a schema, e.g.,
 - Instagram posts - can **vary in the number** of images
 - company reports - can **differ substantially in length**

APPLICATION:

COLLECTING COMPANY SUSTAINABILITY REPORTS

- What are the challenges (and solutions)?

HUMAN VERIFICATION TRIGGERED BY BOT DETECTION

- blocks access to a website completely

Let's confirm you are
human

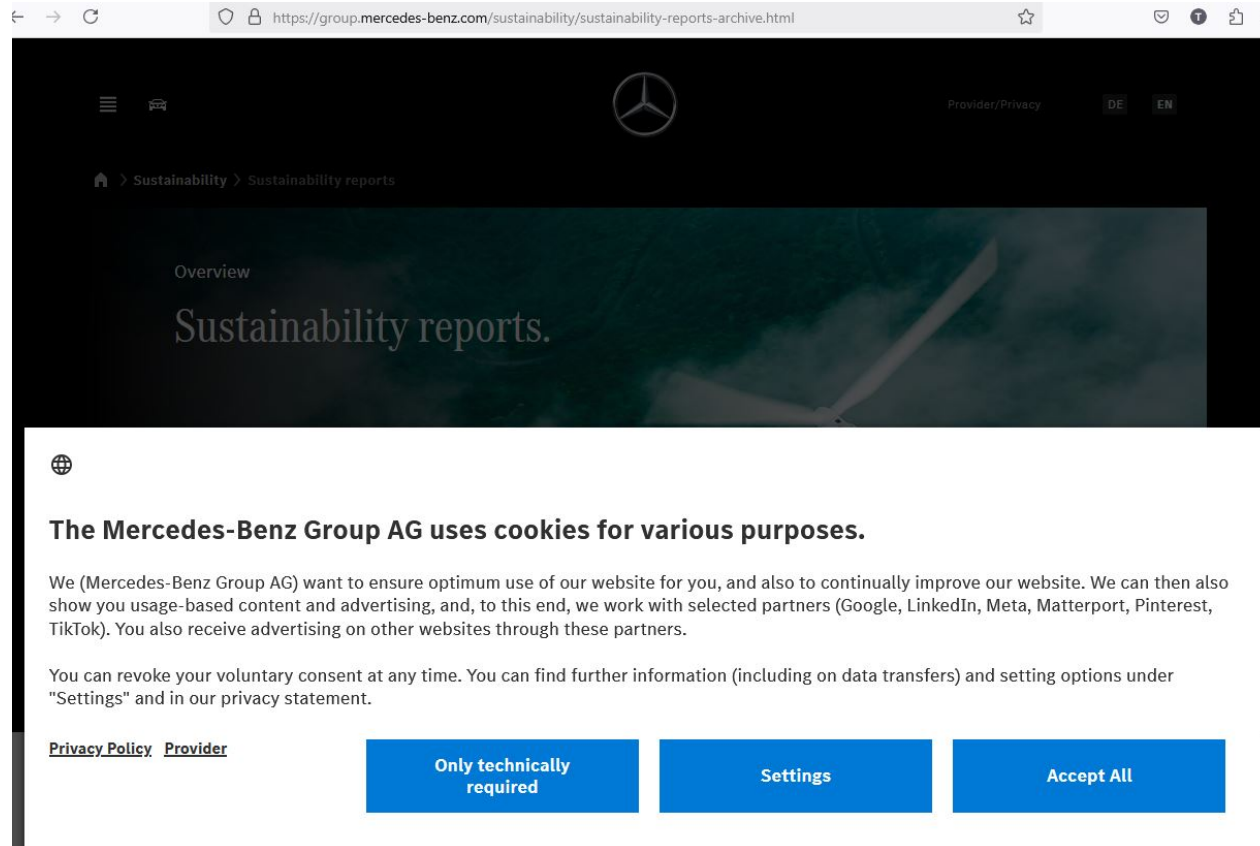
Complete the security check before
continuing. This step verifies that you are not
a bot, which helps to protect your account
and prevent spam.

Begin >

English ▼

COOKIES POPUP

- blocks access to the information on a webpage



OTHER CHALLENGES

- Location of target data
 - often *unknown in advance*
- Downloading multiple data files
 - a tedious task that may lead to *omissions* if done manually
- Extracting texts from pdf files for analysis
 - might need to use an **optical character recognition (OCR)** approach if a file is *not text-searchable*

LET'S DIVE IN

- Week 6 & 7 marimo notebooks

